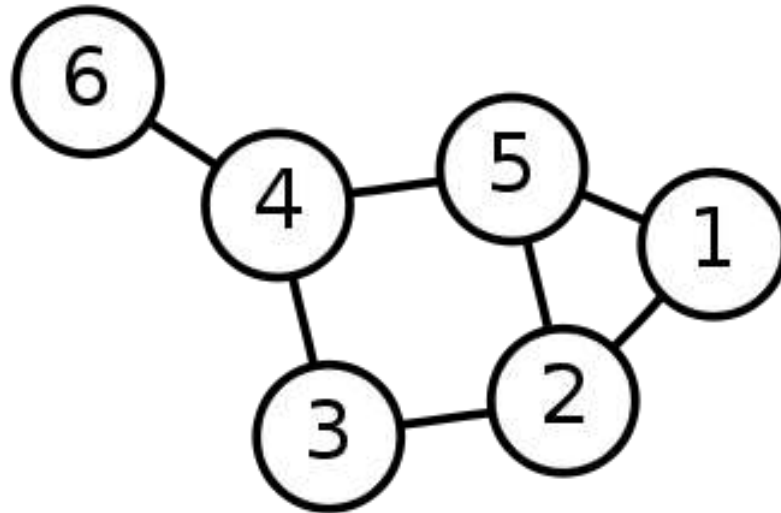


Dijkstra Algoritması

Single-Source Shortest Path Problem

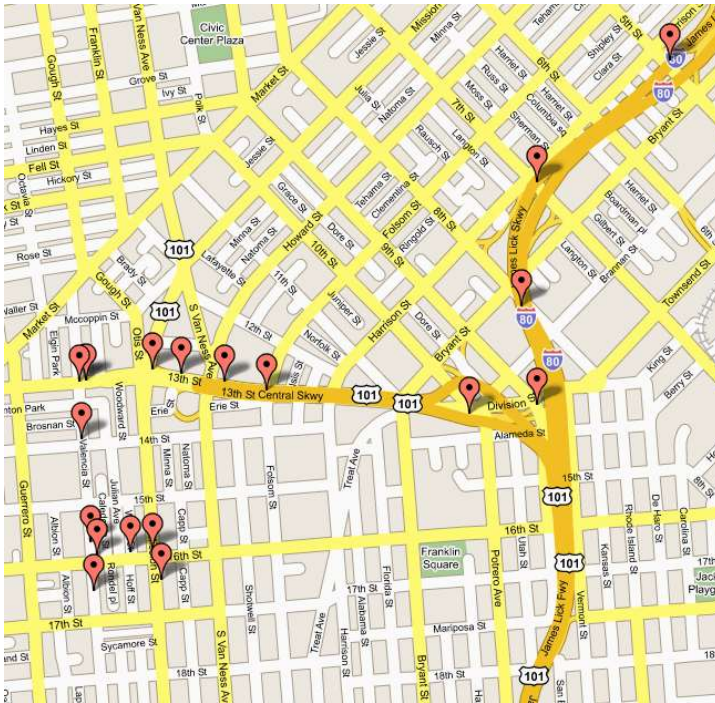
Single-Source Shortest Path Problem - Bir V kaynak tepe noktasından graftaki diğer tüm düğümlere en kısa yolları bulma problemi.



Uygulamalar

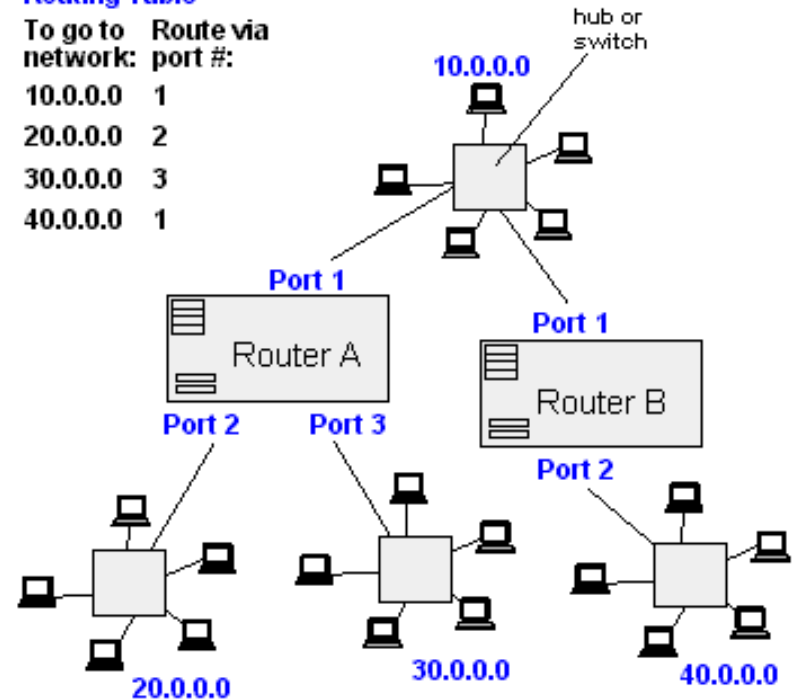
- Maps (Map Quest, Google Maps)
- Yönlendirme Sistemleri (Routing Systems)

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Router A Routing Table

To go to network:	Route via port #:
10.0.0.0	1
20.0.0.0	2
30.0.0.0	3
40.0.0.0	1



Dijkstra's algorithm

Dijkstra's algorithm - Graf teorisindeki tek kaynaklı en kısa yol problemine bir çözümdür..

Hem yönlendirilmiş hem de yönlendirilmemiş grafikler üzerinde çalışır. Ancak, tüm kenarların negatif olmayan ağırlıkları olmalıdır.

Input: $G = \{E, V\}$ Ağırlıklı grafi ve $v \in V$ düğümü, öyle ki tüm kenar ağırlıkları pozitif.

Output: Verilen bir düğümden diğer bütün düğümlere en kısa yol uzunlukları

Dijkstra pseudocode

Dijkstra(v1, v2):

for each vertex v: // Initialization

v's distance := infinity.

v's previous := none.

v1's distance := 0.

List := {all vertices}.

while List is not empty:

v := remove List vertex with minimum distance.

mark v as known.

for each unknown neighbor n of v:

dist := v's distance + edge (v, n)'s weight.

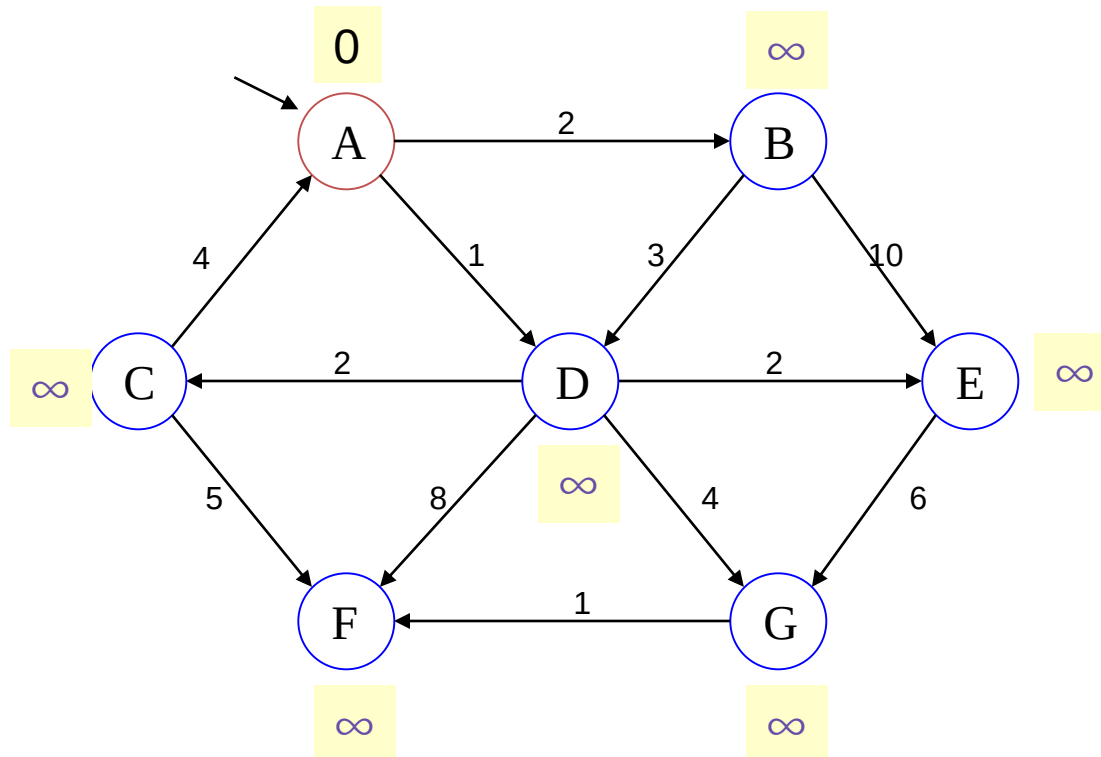
if dist is smaller than n's distance:

n's distance := dist.

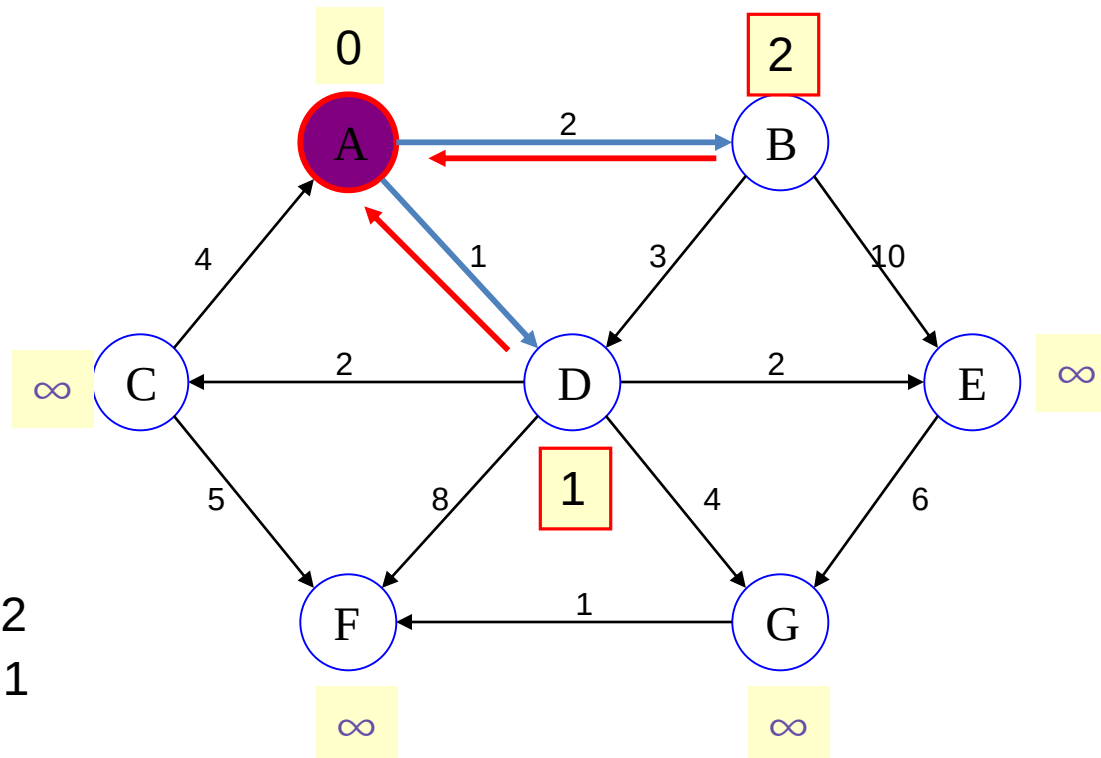
n's previous := v.

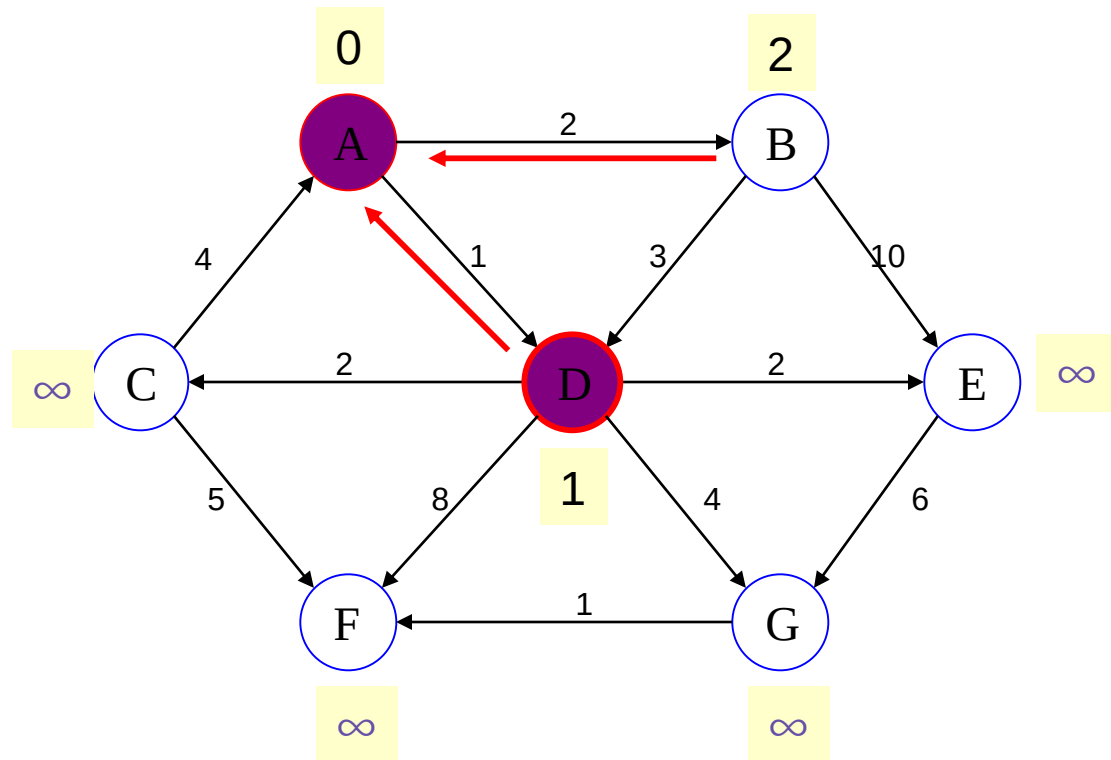
*reconstruct path from v2 back to v1,
following previous pointers.*

Örnek

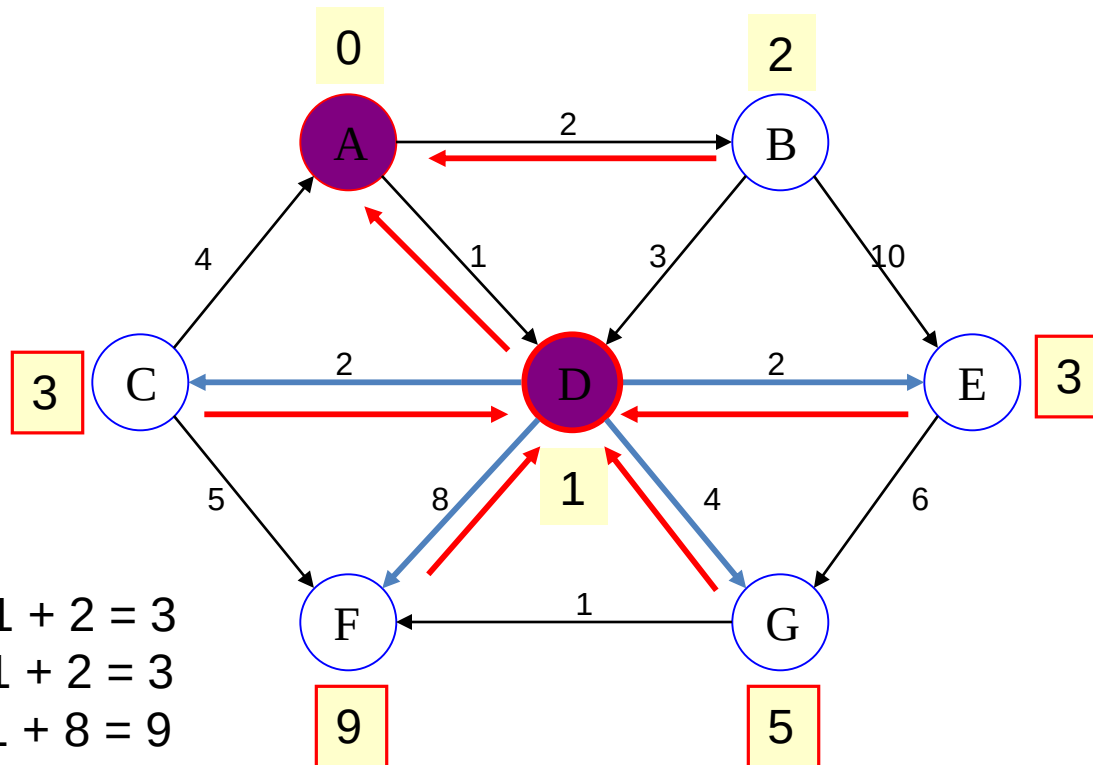


En yakın komşulukların düzenlenmesi





Komşulukların Güncellenmesi

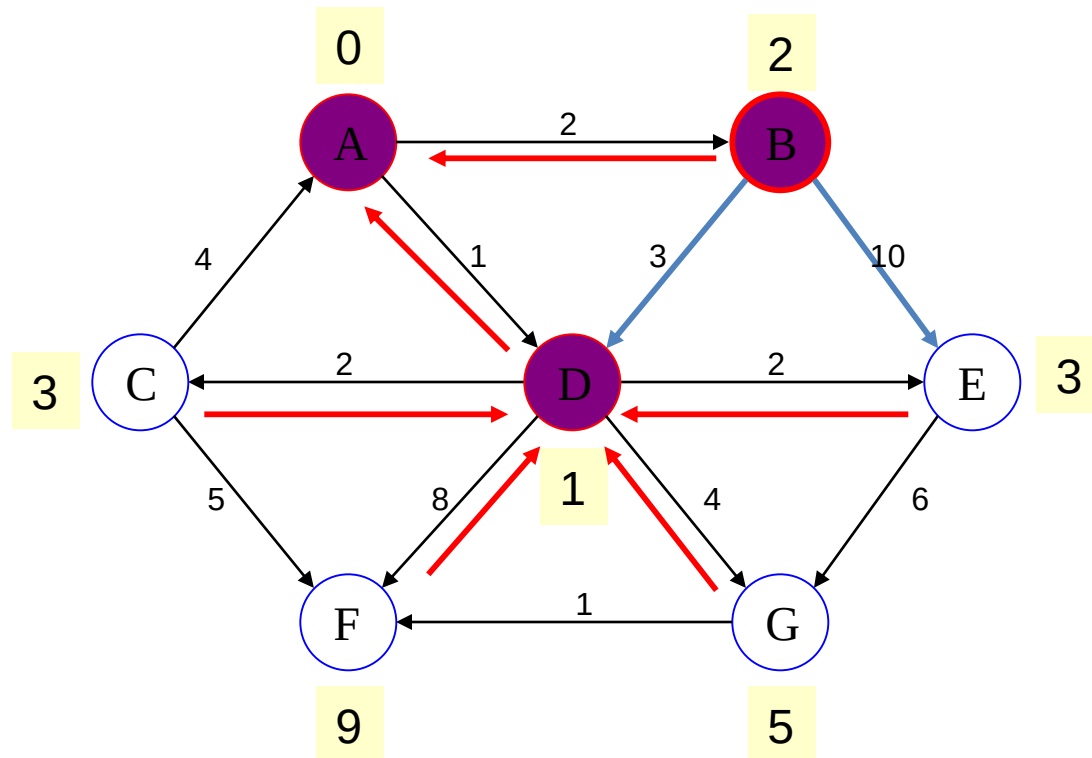


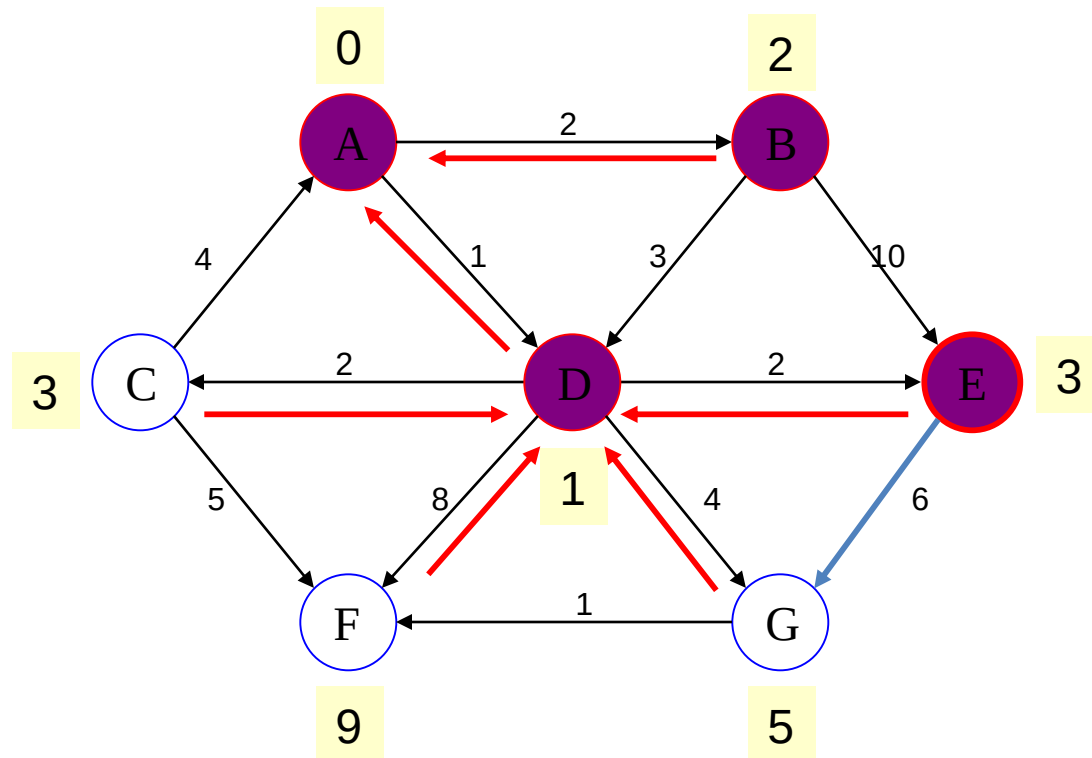
$$\text{Distance}(C) = 1 + 2 = 3$$

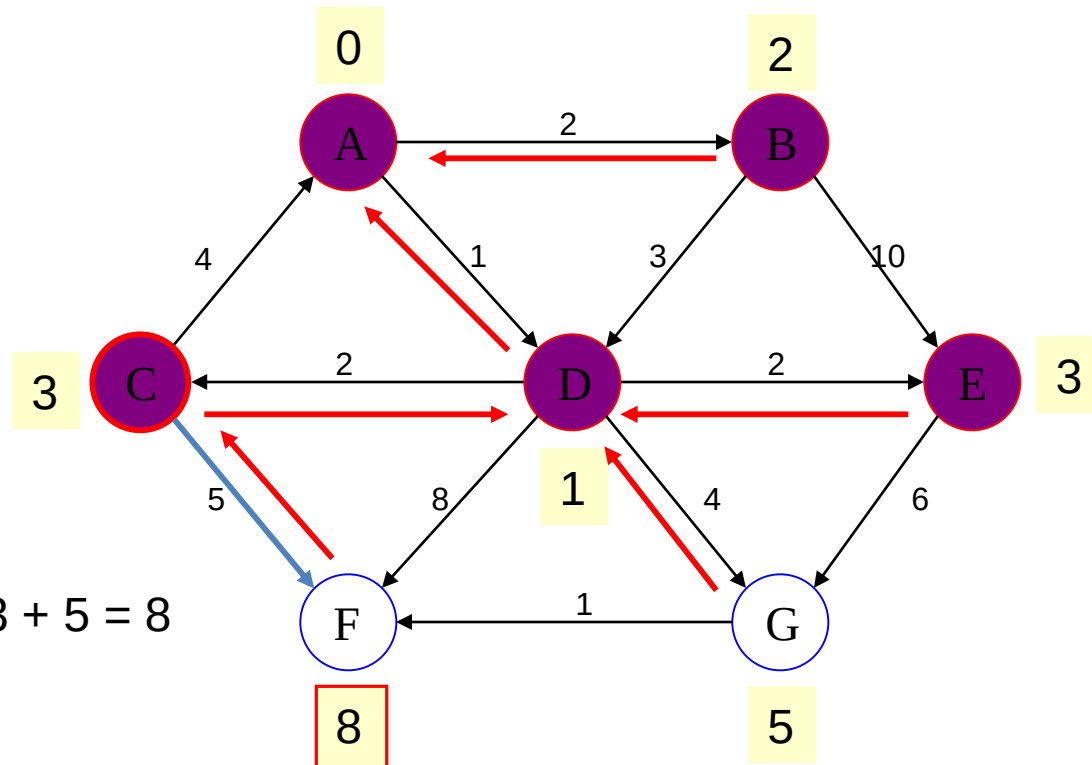
$$\text{Distance}(E) = 1 + 2 = 3$$

$$\text{Distance}(F) = 1 + 8 = 9$$

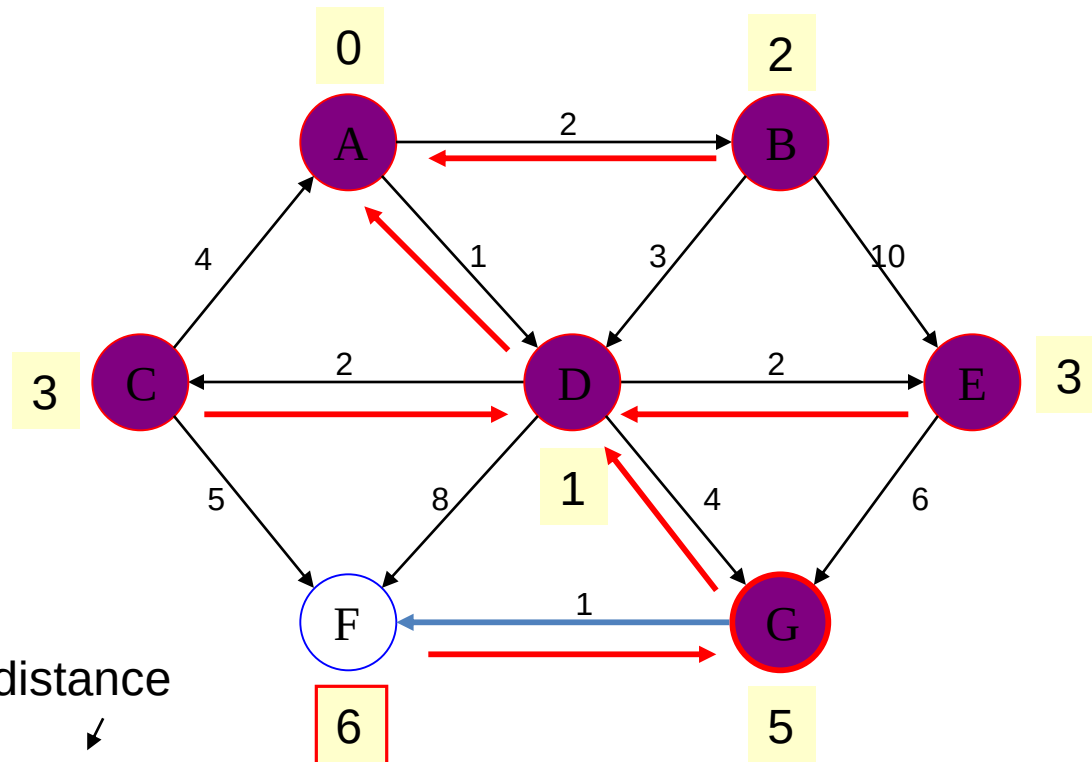
$$\text{Distance}(G) = 1 + 4 = 5$$







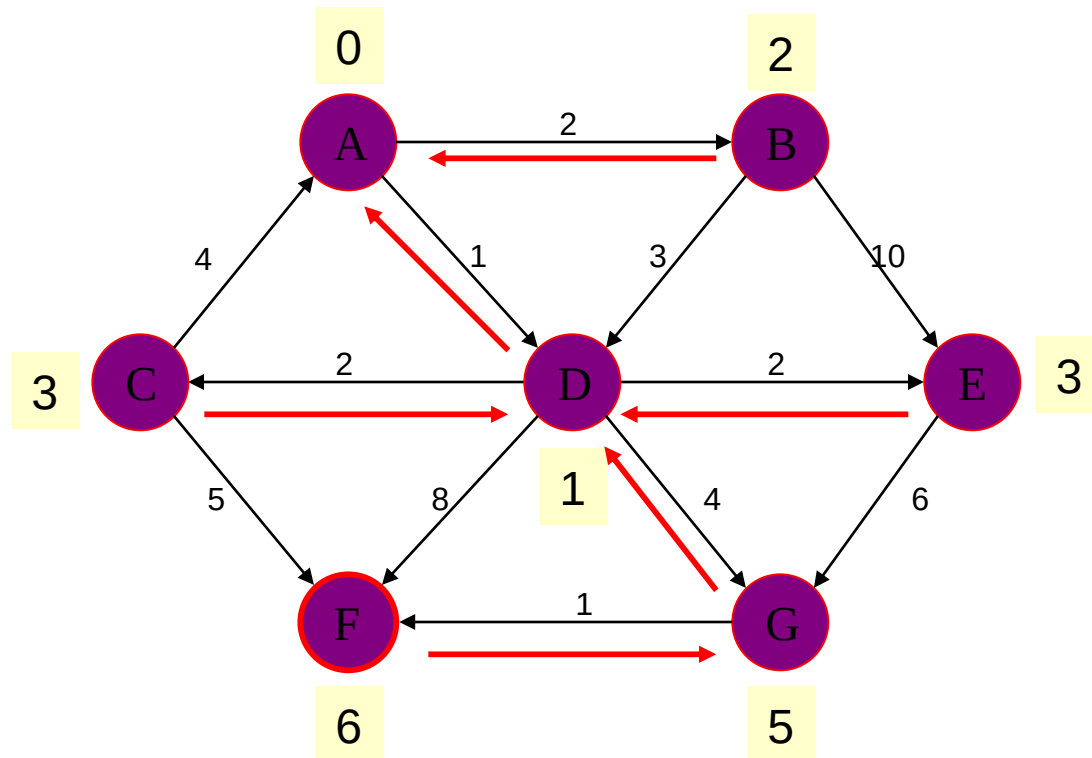
Distance(F) = 3 + 5 = 8



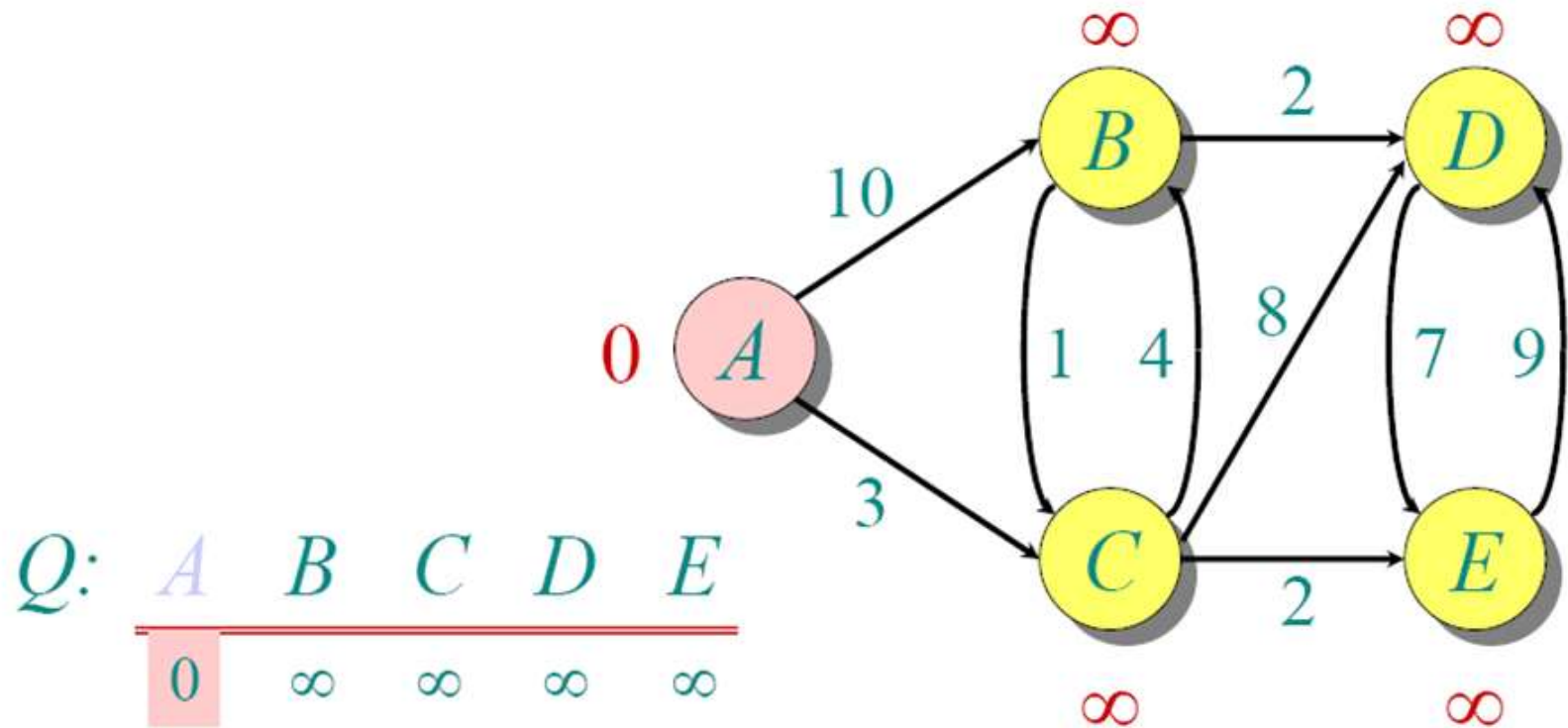
Previous distance



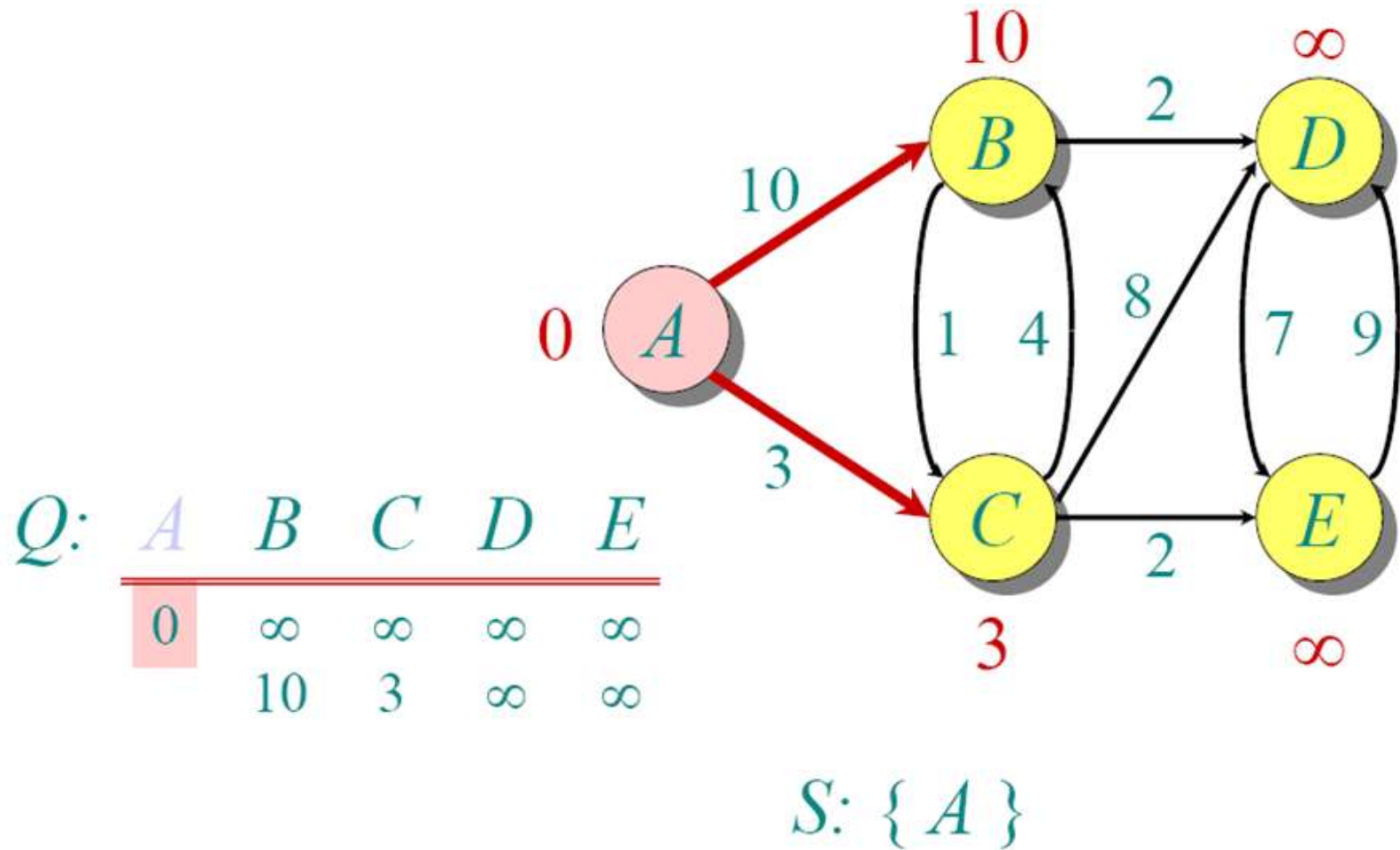
$$\text{Distance}(F) = \min (8, 5+1) = 6$$



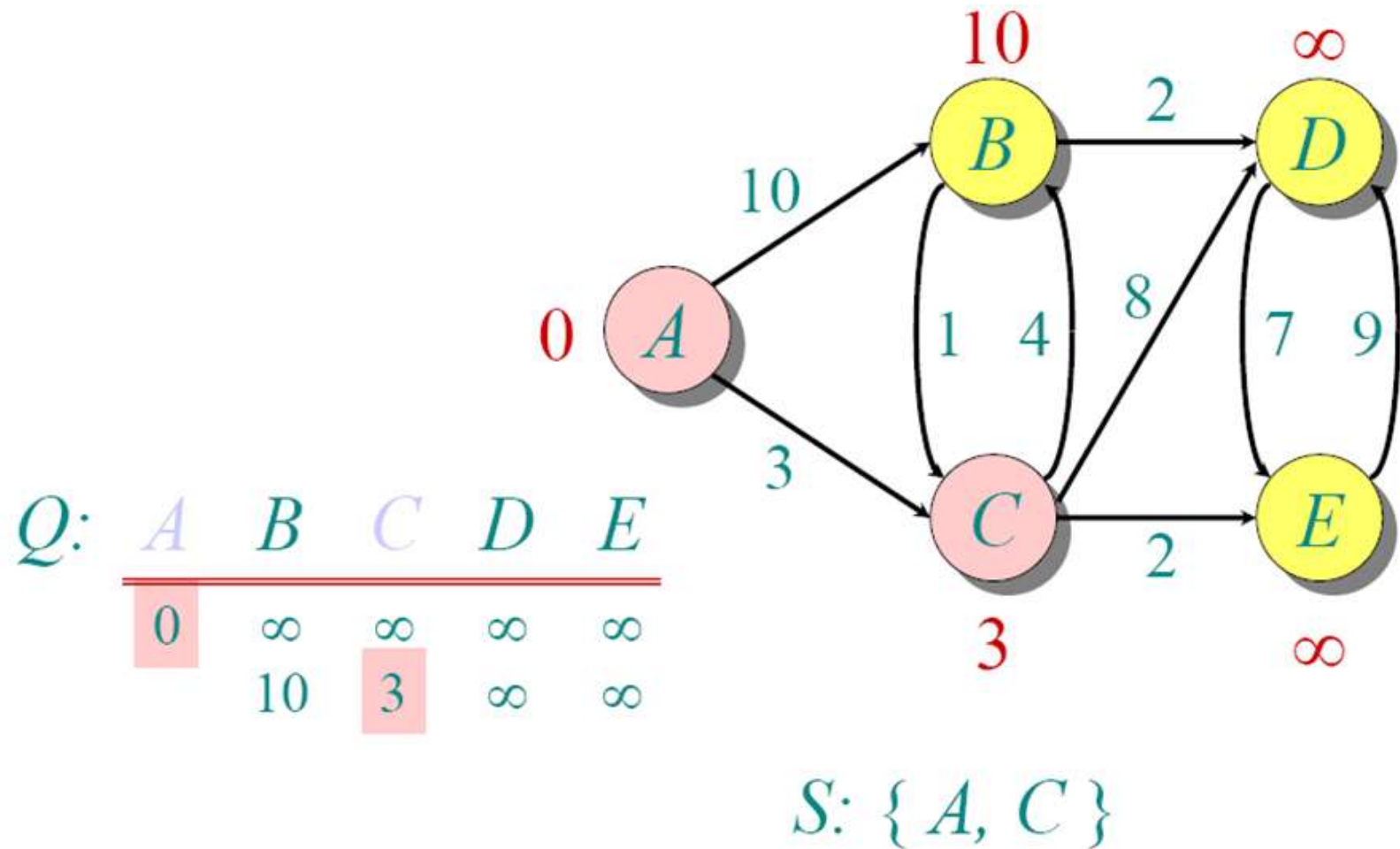
Örnek 2



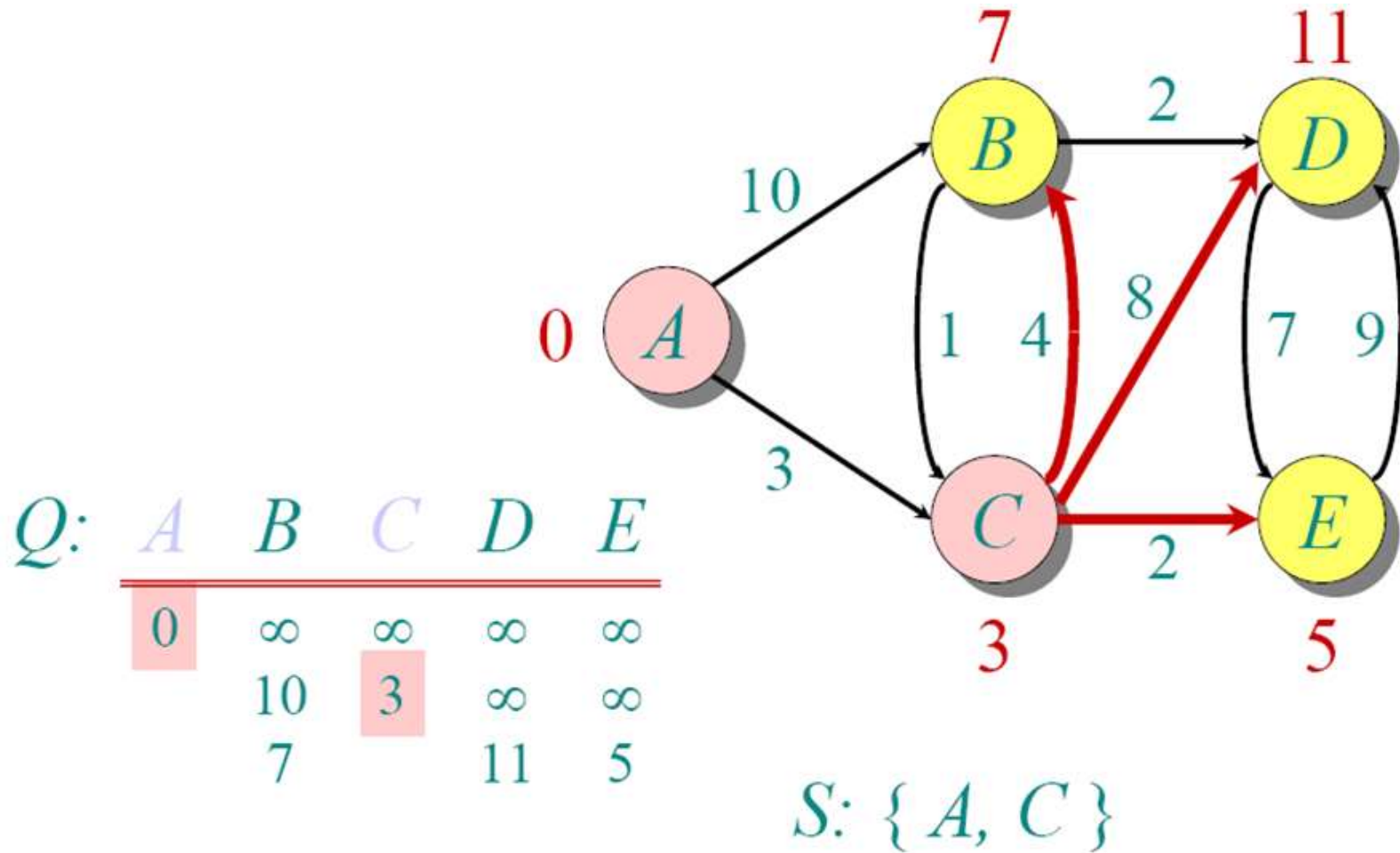
Örnek 2 Devam



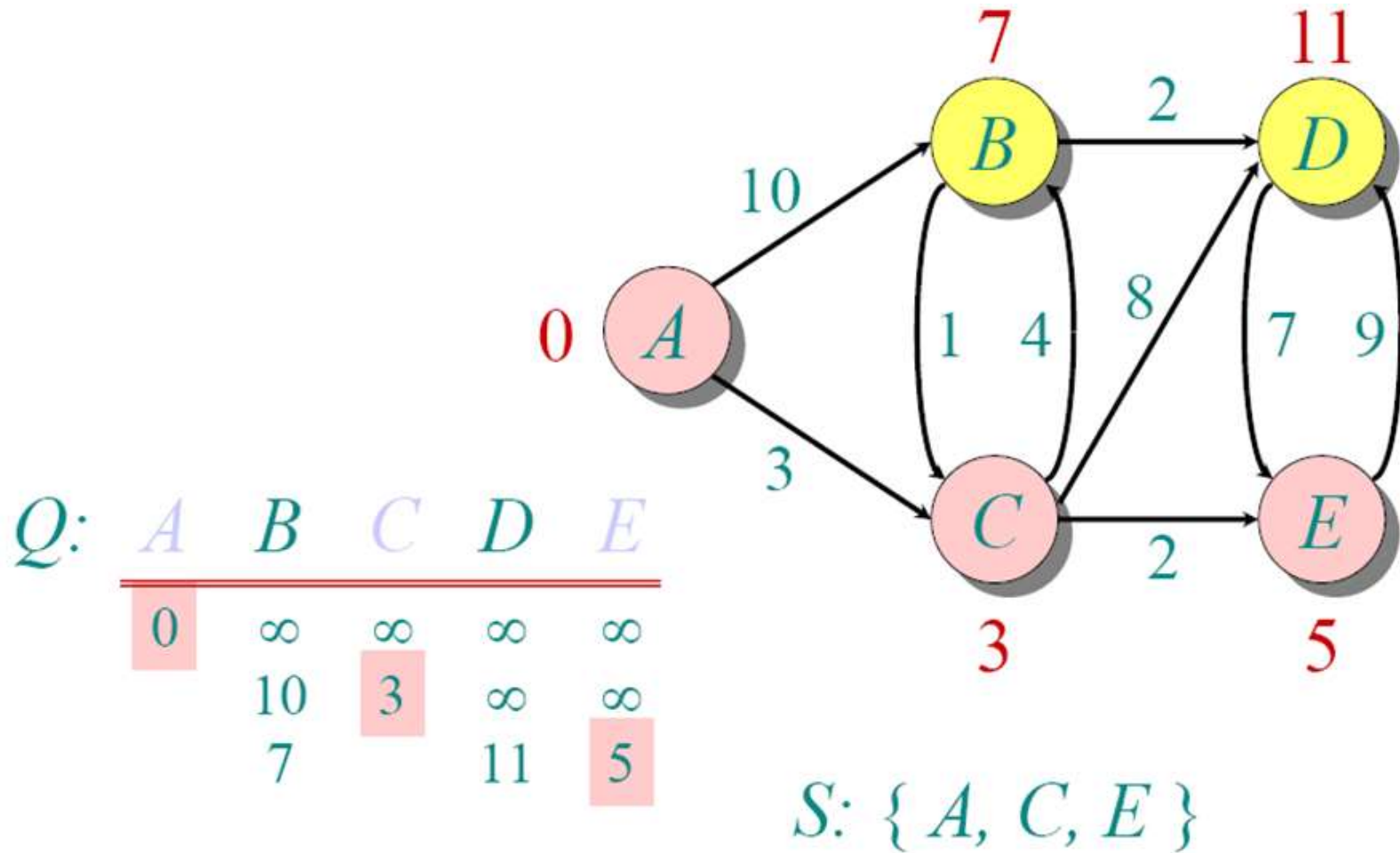
Örnek 2 Devam



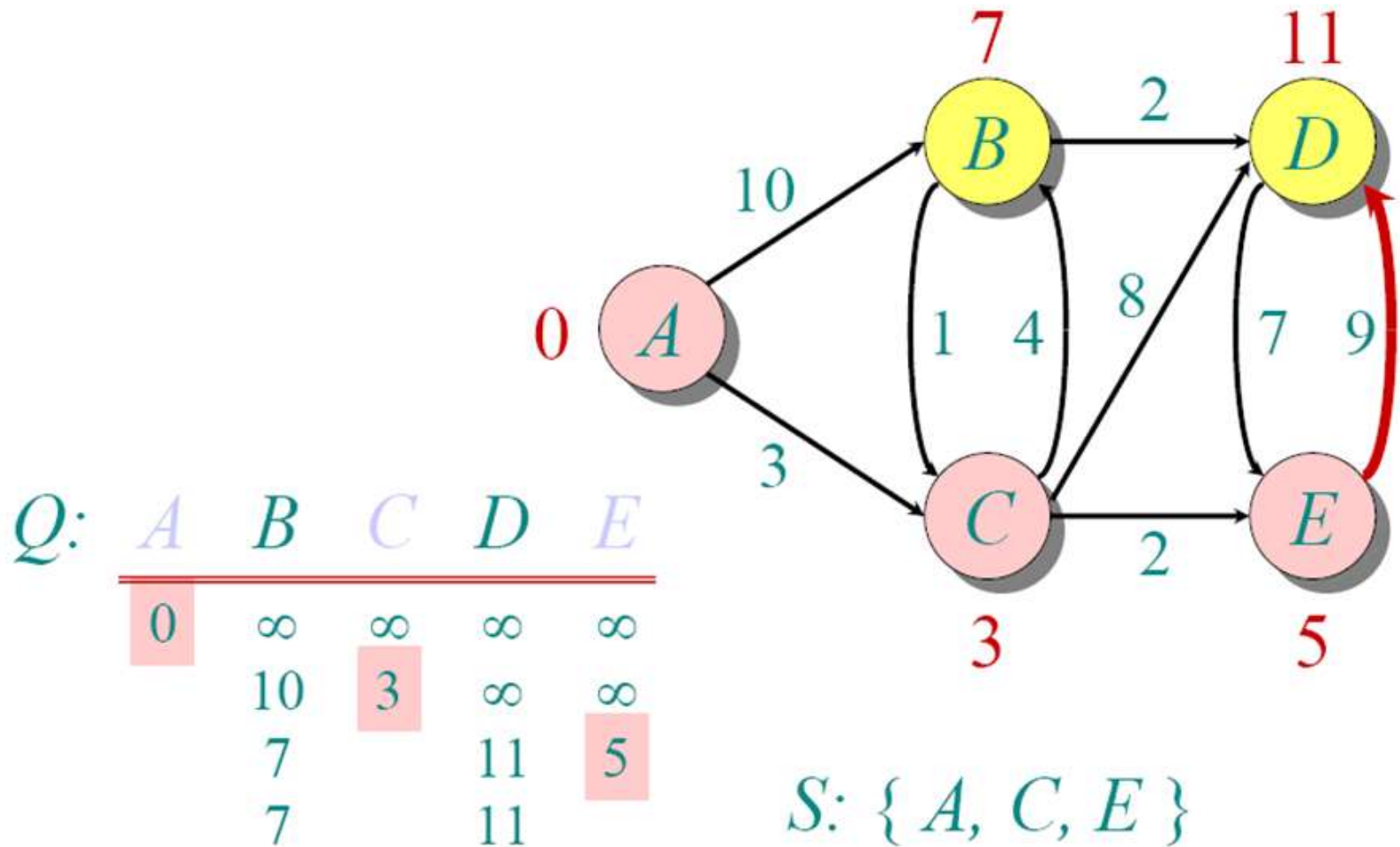
Örnek 2 Devam



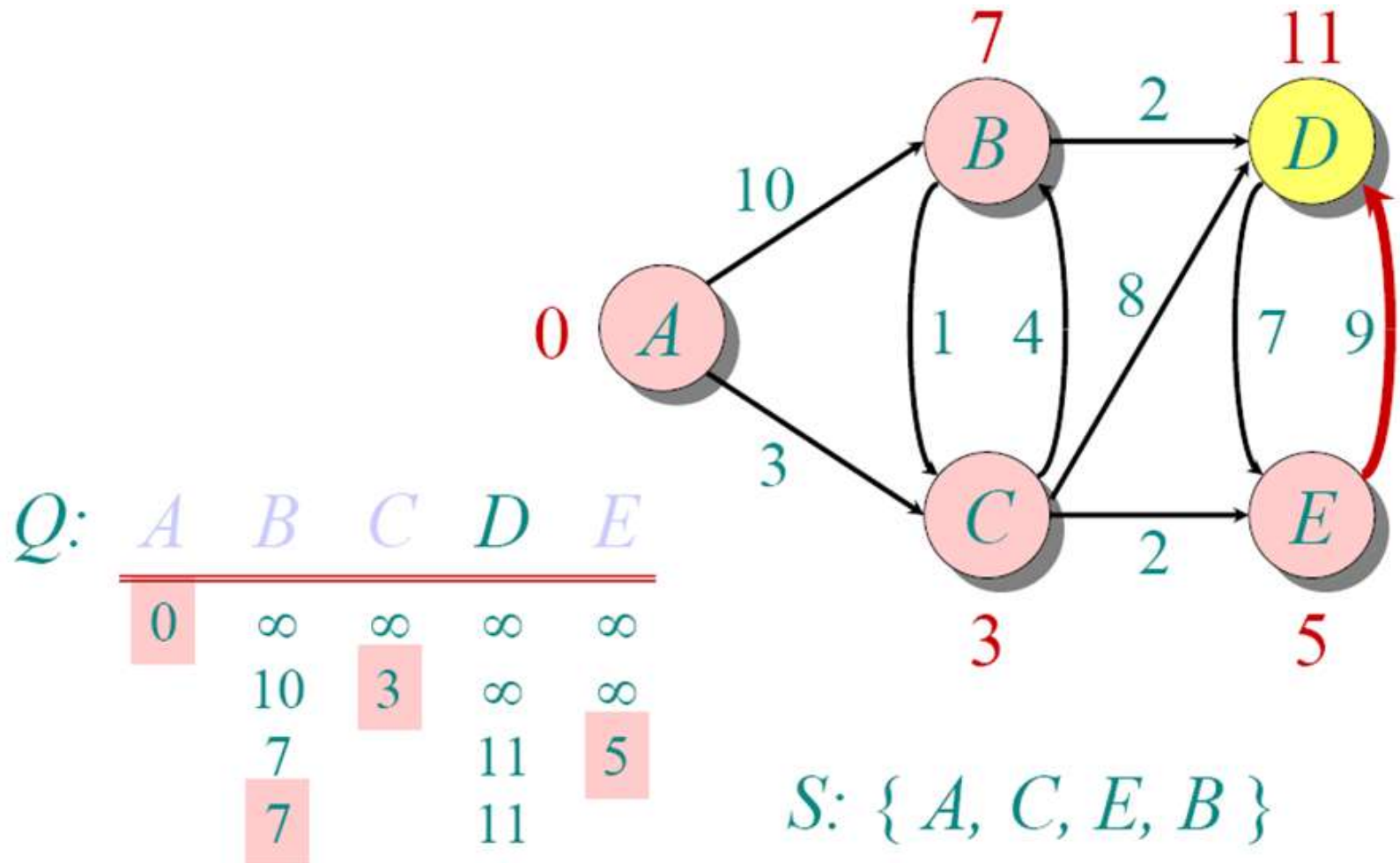
Örnek 2 Devam



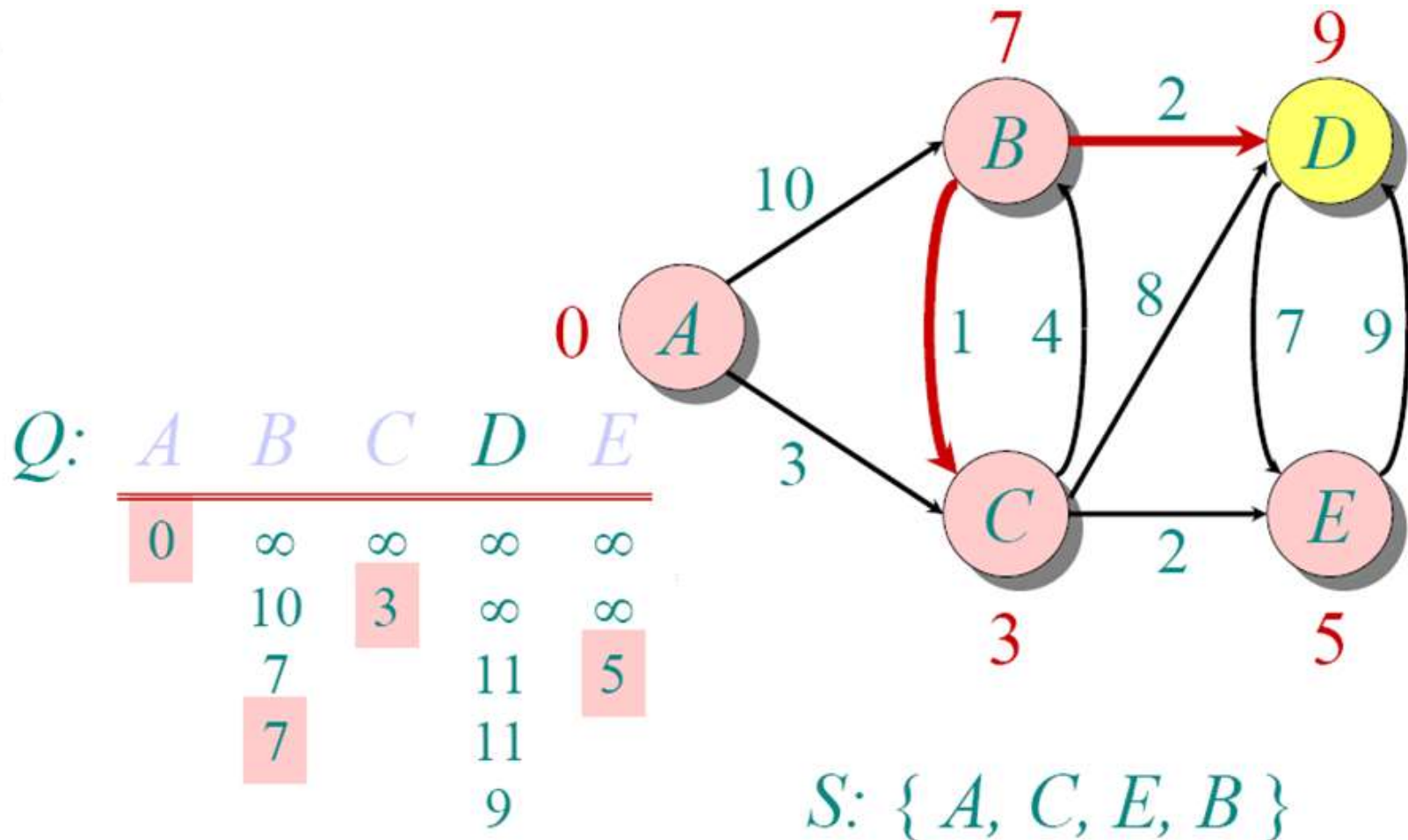
Örnek 2 Devam



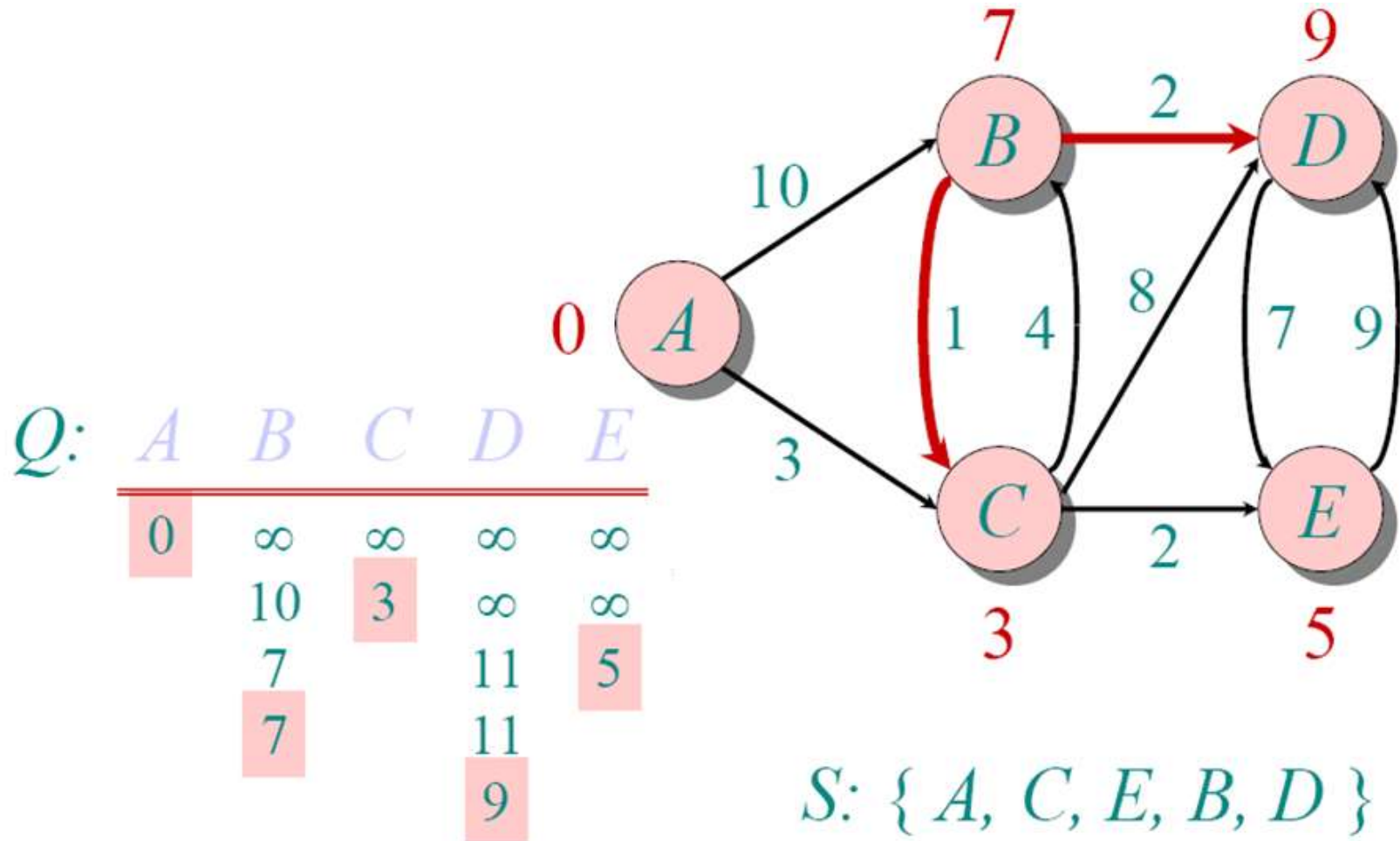
Örnek 2 Devam



Örnek 2 Devam



Örnek 2 Devam



Dijkstra's Algoritması

- Graph G , weight function w , root s

DIJKSTRA(G, w, s)

1 **for** each $v \in V$

2 **do** $d[v] \leftarrow \infty$

3 $d[s] \leftarrow 0$

4 $S \leftarrow \emptyset$ $\triangleright$ Set of discovered nodes

5 $Q \leftarrow V$

6 **while** $Q \neq \emptyset$

7 **do** $u \leftarrow \text{EXTRACT-MIN}(Q)$

8 $S \leftarrow S \cup \{u\}$

9 **for** each $v \in \text{Adj}[u]$

10 **do if** $d[v] > d[u] + w(u, v)$

11 **then** $d[v] \leftarrow d[u] + w(u, v)$

relaxing
edges

Zaman Karmaşıklığı: Bağlı Liste

Dijkstra algoritmasının en basit uygulaması, köşeleri sıradan bir bağlantılı liste veya dizide depolamaktır.

- $|V|$ düğümler ve $|E|$ kenarlar
- Başlangıç $O(|V|)$
- While loop $O(|V|)$
 - minimum ağırlıklı düğümü Bul ve kaldır $O(|V|)$
- Olası güncelleme sayısı : $|E|$
 - Güncelleme maliyeti $O(1)$

Toplam zaman maliyeti: $O(|V^2| + |E|) = O(|V^2|)$

Zaman Karmaşıklığı: Öncelikli Kuyruk

- Heapleştirme kullanılacak: $O(|V|)$
- While loop $O(|V|)$
 - minimum ağırlıklı düğümü Bul ve kaldır $O(\log |V|)$ ve $O(\log |V|)$ deleteMin
- Olası güncelleme sayısı: $|E|$
 - Güncelleme Maliyeti $O(\log |V|)$

Toplam Zaman Maliyeti: $O(|V|\log|V| + |E|\log|V|) = O(|E|\log|V|)$

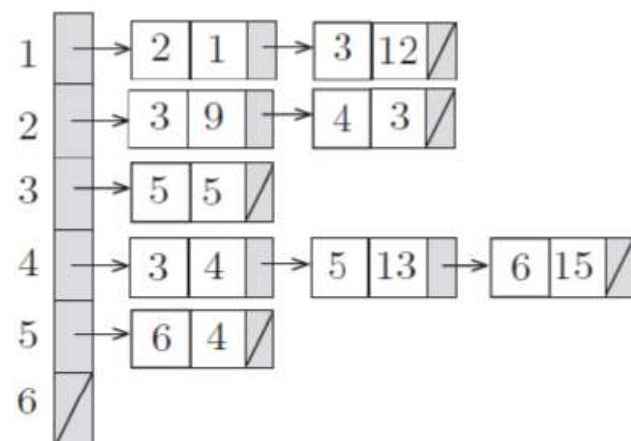
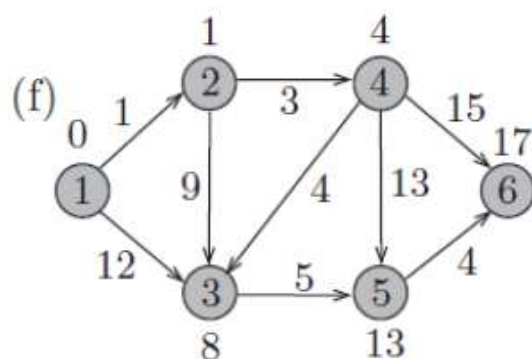
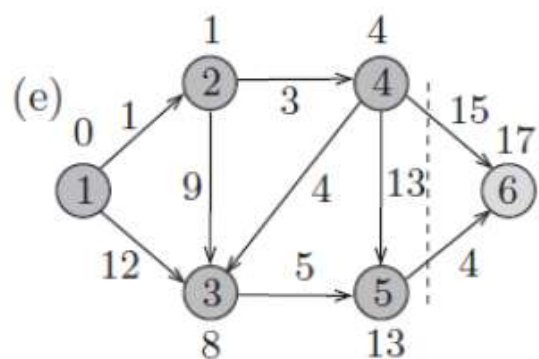
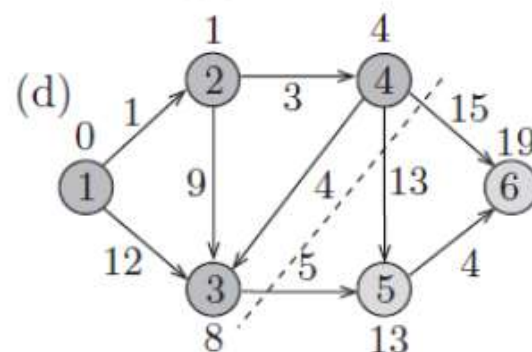
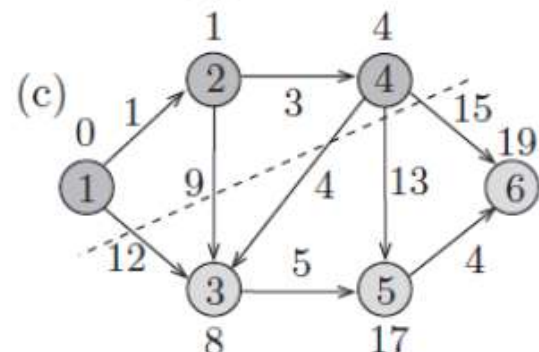
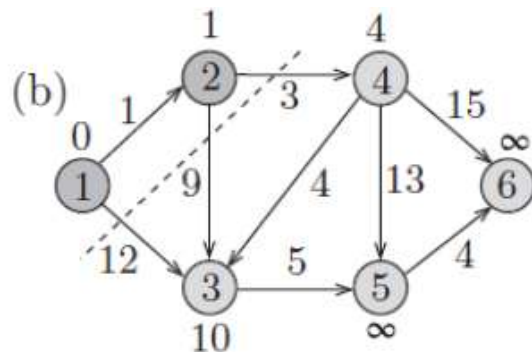
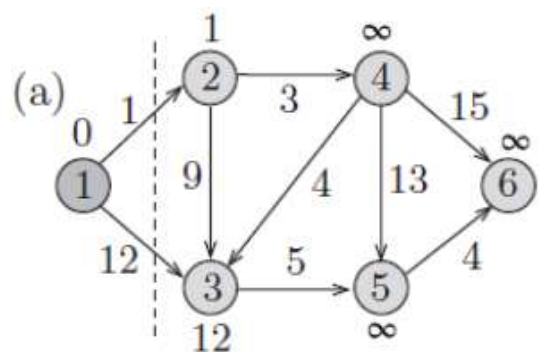
- $|V| = O(|E|)$ (Bağılantılı graf varsayımı ile)

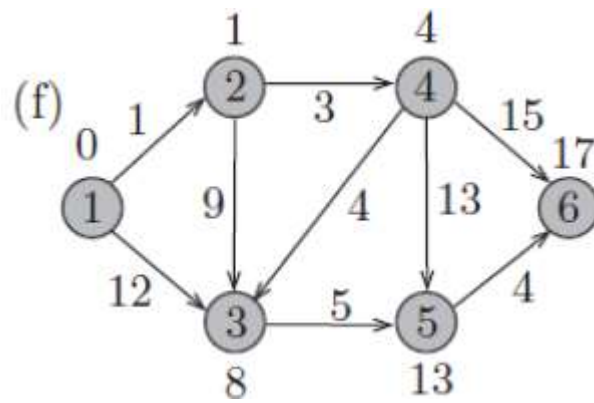
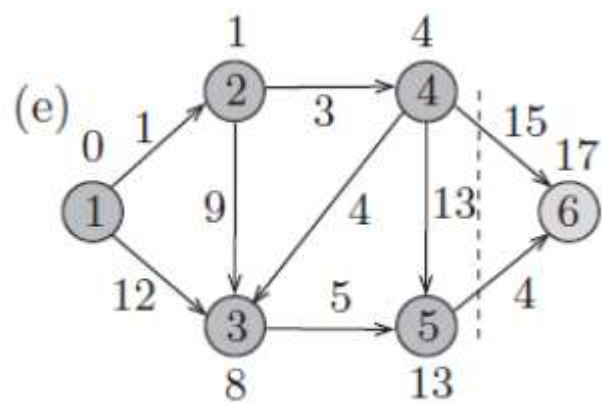
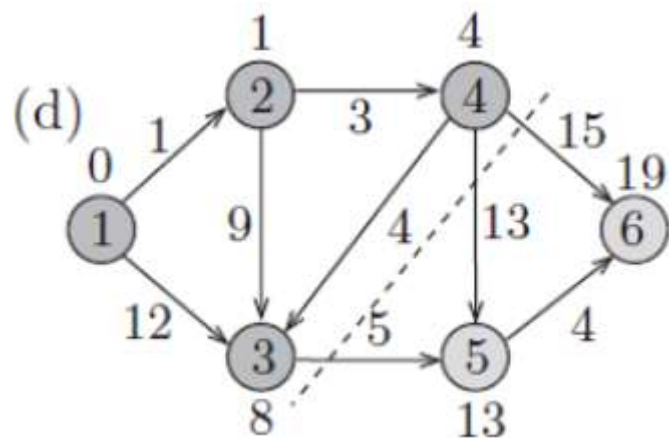
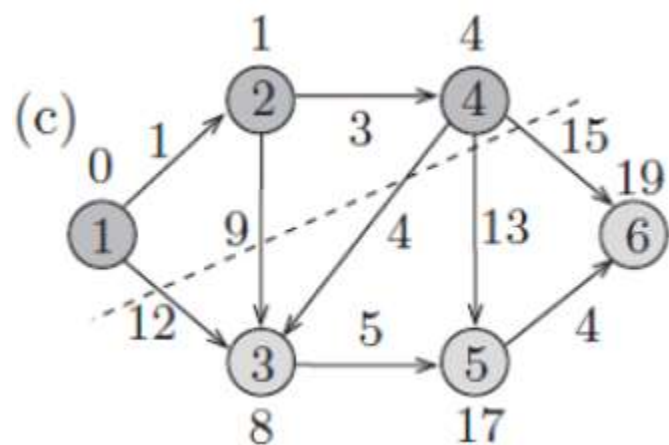
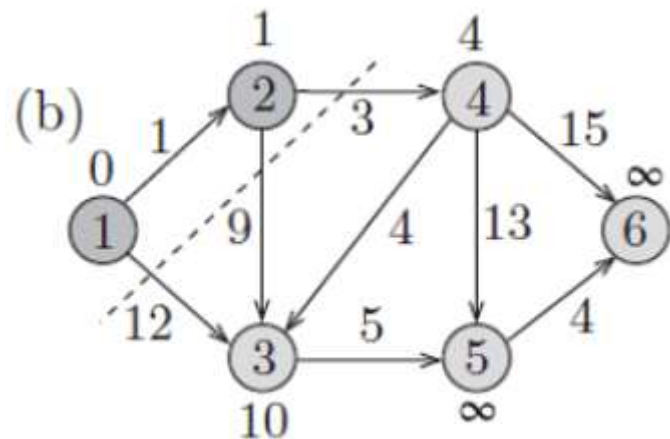
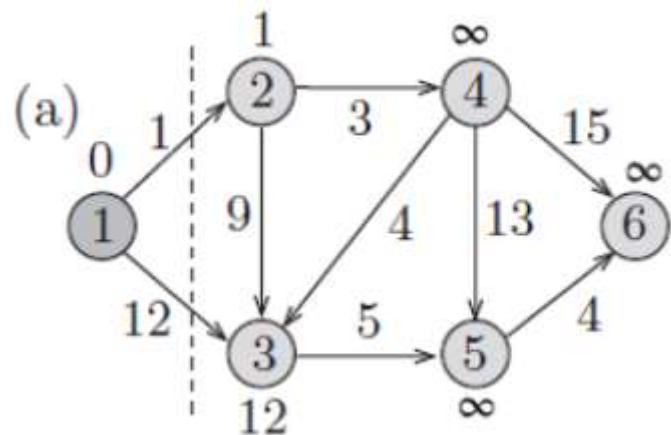
Algorithm DIJKSTRA

Input: $G = (V, E)$, $V = \{1, 2, \dots, n\}$.

Output: The distance from vertex 1 to every other vertex in G .

1. $X = \{1\}$; $Y \leftarrow V - \{1\}$; $\lambda[1] \leftarrow 0$
2. for $y \leftarrow 2$ to n
3. if y is adjacent to 1 then $\lambda[y] \leftarrow \text{length}[1, y]$
4. else $\lambda[y] \leftarrow \infty$
5. end if
6. end for
7. for $j \leftarrow 1$ to $n - 1$
8. Let $y \in Y$ be such that $\lambda[y]$ is minimum
9. $X \leftarrow X \cup \{y\}$ {add vertex y to X }
10. $Y \leftarrow Y - \{y\}$ {delete vertex y from Y }
11. for each edge (y, w)
12. if $w \in Y$ and $\lambda[y] + \text{length}[y, w] < \lambda[w]$ then
13. $\lambda[w] \leftarrow \lambda[y] + \text{length}[y, w]$
14. end if
15. end for





Dinamik Programlama

Floyd Warshall Algoritması

$$d_{i,j}^k = \begin{cases} l(i,j) & \text{if } k=0 \\ \min \{ d_{i,j}^{k-1}, d_{i,k}^{k-1} + d_{k,j}^{k-1} \} & \text{if } 1 \leq k \leq n \end{cases}$$

$d_{i,j}^k$: i 'den j 'ye en kısa yolun uzunluğudur.

(Bu yol $k+1, k+2, \dots, n$.
düğümlerden geçmez.)

Örneğin $d_{i,j}^2$, i 'den j 'ye en kısa yolu gösterir. Bu yol 1 ve 2'den başka düğümlerden geçmez.

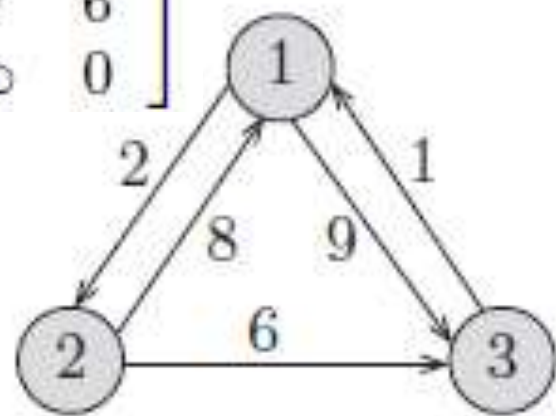
$$k=1$$

$$D(1,1) = \min \{ D(1,1), D(1,1) + D(1,1) \} = 0$$

$$D(1,2) = \min \{ D(1,2), D(1,1) + D(1,2) \} = 2$$

$$D(1,3) = \min \{ D(1,3), D(1,1) + D(1,3) \} = 9$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix}$$



$$\textcircled{i=2} \quad D(2,1) = \min \{ D(2,1), D(2,1) + D(1,1) \} = 8$$

$$D(2,2) = \min \{ D(2,2), D(2,1) + D(1,2) \} = 0$$

$$D(2,3) = \min \{ D(2,3), D(2,1) + D(1,3) \} = 6$$

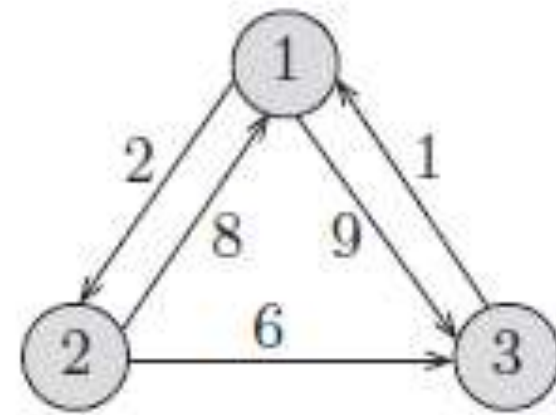
$$D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$\textcircled{i=3} \quad D(3,1) = \min \{ D(3,1), D(3,1) + D(1,1) \} = 1$$

$$D(3,2) = \min \{ D(3,2), D(3,1) + D(1,2) \} = \textcircled{3}$$

$$D(3,3) = \min \{ D(3,3), D(3,1) + D(1,3) \} = 0$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix}$$



$$k=1$$

$$i=1 \quad D(1,1) = \min \{ D(1,1), D(1,1) + D(1,1) \} = 0$$

$$D(1,2) = \min \{ D(1,2), D(1,1) + D(1,2) \} = 2$$

$$D(1,3) = \min \{ D(1,3), D(1,1) + D(1,3) \} = 9$$

$$i=2 \quad D(2,1) = \min \{ D(2,1), D(2,1) + D(1,1) \} = 8$$

$$D(2,2) = \min \{ D(2,2), D(2,1) + D(1,2) \} = 0$$

$$D(2,3) = \min \{ D(2,3), D(2,1) + D(1,3) \} = 6$$

$$i=3 \quad D(3,1) = \min \{ D(3,1), D(3,1) + D(1,1) \} = 1$$

$$D(3,2) = \min \{ D(3,2), D(3,1) + D(1,2) \} = 3$$

$$D(3,3) = \min \{ D(3,3), D(3,1) + D(1,3) \} = 0$$

$$D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$k=2, \textcircled{i=1}, D(1,1) = \min \{D(1,1), D(1,2) + D(2,1)\} = 0$$

$$D(1,2) = \min \{D(1,2), D(1,2) + D(2,2)\} = 2$$

$$D(1,3) = \min \{D(1,3), D(1,2) + D(2,3)\} = \textcircled{8}$$

$$\textcircled{i=2}, D(2,1) = \min \{D(2,1), D(2,2) + D(2,1)\} = 8$$

$$D(2,2) = \min \{D(2,2), D(2,2) + D(2,2)\} = 0$$

$$D(2,3) = \min \{D(2,3), D(2,2) + D(2,3)\} = 6$$

$$\textcircled{i=3}, D(3,1) = \min \{D(3,1), D(3,2) + D(2,1)\} = 1$$

$$D(3,2) = \min \{D(3,2), D(3,2) + D(2,2)\} = 3$$

$$D(3,3) = \min \{D(3,3), D(3,2) + D(2,3)\} = 0$$

$$D_2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$k=3 \quad i=1 \quad D(1,1) = \min \{D(1,1), D(1,3) + D(3,1)\} = 0$$

$$D(1,2) = \min \{D(1,2), D(1,3) + D(3,2)\} = 2$$

$$D(1,3) = \min \{D(1,3), D(1,3) + D(3,3)\} = 8$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix} \quad D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix} \quad D_2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$\textcircled{i=2} \quad D(2,1) = \min \{D(2,1), D(2,3) + D(3,1)\} = 7$$

$$D(2,2) = \min \{D(2,2), D(2,3) + D(3,2)\} = 0$$

$$\underline{D(2,3) = \min \{D(2,3), D(2,3) + D(3,3)\} = 6}$$

$$D_3 = \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$\textcircled{i=3} \quad D(3,1) = \min \{D(3,1), D(3,3) + D(3,1)\} = 1$$

$$D(3,2) = \min \{D(3,2), D(3,3) + D(3,2)\} = 3$$

$$D(3,3) = \min \{D(3,3), D(3,3) + D(3,3)\} = 0$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D_2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$k=3 \quad i=1$

$$D(1,1) = \min \{ D(1,1), D(1,3) + D(3,1) \} = 0$$

$$D(1,2) = \min \{ D(1,2), D(1,3) + D(3,2) \} = 2$$

$$D(1,3) = \min \{ D(1,3), D(1,3) + D(3,3) \} = 8$$

$i=2$

$$D(2,1) = \min \{ D(2,1), D(2,3) + D(3,1) \} = 7$$

$$D(2,2) = \min \{ D(2,2), D(2,3) + D(3,2) \} = 0$$

$$\underline{D(2,3) = \min \{ D(2,3), D(2,3) + D(3,3) \} = 6}$$

$i=3$

$$D(3,1) = \min \{ D(3,1), D(3,3) + D(3,1) \} = 1$$

$$D(3,2) = \min \{ D(3,2), D(3,3) + D(3,2) \} = 3$$

$$D(3,3) = \min \{ D(3,3), D(3,3) + D(3,3) \} = 0$$

$$D_3 = \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

D_3 matrisi aranan matristir. Görüldüğü gibi üçüncü (k) iterasyonda 3. satır ve 3. sütun değişmemiştir.

$$D_3 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix} \end{matrix}$$

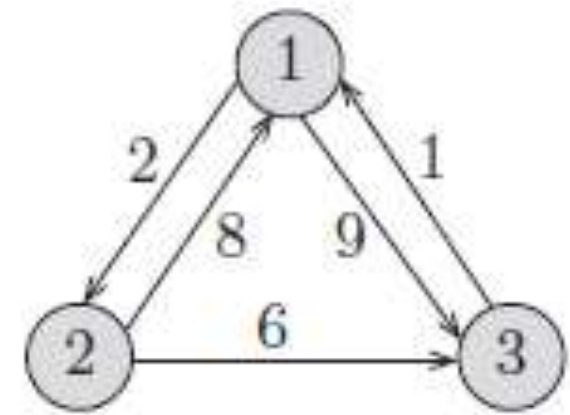
1. düğüm ile 3. düğüm arasındaki en kısa yol 8 birimdir.

3. düğüm ile 2. düğüm arasındaki en kısa yol 2 birimdir

2. düğüm ile 3. düğüm arasındaki en kısa yol 6 birimdir.

Her zaman k. iterasyonda; k. sütun ile k. satır değişmeden kalır.

$$d_{i,j}^k = \begin{cases} l[i,j] & \text{if } k = 0 \\ \min\{d_{i,j}^{k-1}, d_{i,k}^{k-1} + d_{k,j}^{k-1}\} & \text{if } 1 \leq k \leq n. \end{cases}$$



$$D_k[i,j] = \min\{D_{k-1}[i,j], D_{k-1}[i,k] + D_{k-1}[k,j]\}.$$

$$D_0 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & \infty & 0 \end{bmatrix}$$

$$D_1 = \begin{bmatrix} 0 & 2 & 9 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D_2 = \begin{bmatrix} 0 & 2 & 8 \\ 8 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}$$

$$D_3 = \begin{bmatrix} 0 & 2 & 8 \\ 7 & 0 & 6 \\ 1 & 3 & 0 \end{bmatrix}.$$

Algorithm FLOYD

Input: $n \times n$ matrix $l[1..n, 1..n]$ $l[i,j]$: graftaki i,j kenarının uzunluğu

Output: $D[i,j]$ = i 'den j 'ye uzaklık

```

1.  $D \leftarrow l$ 
2. for  $k \leftarrow 1$  to  $n$ 
3.   for  $i \leftarrow 1$  to  $n$ 
4.     for  $j \leftarrow 1$  to  $n$ 
5.        $D[i,j] = \min\{D[i,j], D[i,k] + D[k,j]\}$ 
6.     end for
7.   end for
8. end for

```

Algoritmanın çalışma zamanı: $\Theta(n^3)$

Bellek Maliyeti: $\Theta(n^2)$